P-CHARM:
Phonetic Charade Reasoning Model for LLMs

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Abstract : Large Language Models (LLMs) have recently achieved impressive performance in natural language generation and reasoning tasks, fostering widespread adoption across numerous applications. However, beyond well-studied factual hallucinations, LLMs also suffer from more subtle reasoning failures, where outputs appear coherent and convincing despite being logically or linguistically invalid. In this work, we investigate such limitations through the lens of phonetic reasoning, focusing on *charades*, a class of structured phonetic puzzles that require explicit decomposition and recomposition of speech sounds. We show that current LLMs struggle to generate phonologically coherent charades, revealing a fundamental weakness in sound-level reasoning. We introduce **P-CHARM**, a framework for analyzing and improving phonetic reasoning in LLMs. Our contributions include (i) an empirical analysis of LLM failures on phonetic charades, (ii) a semi-automatically constructed dataset of phonetic charades, and (iii) an evaluation protocol based on phonological feature representations and edit-distance metrics. We further demonstrate that simple prompting strategies combining few-shot examples and chain-of-thought reasoning significantly improve phonetic coherence without modifying model parameters.

1 Introduction

1.1 Context and Motivation

Large Language Models (LLMs) have recently achieved remarkable performance across a wide range of Natural Language Processing tasks, including text generation, problem solving, and logical reasoning. Their ability to produce fluent and structured explanations often creates the impression of strong reasoning capabilities.

However, recent work has highlighted important limitations in these models, extending beyond factual hallucinations to more subtle forms of reasoning failure. In particular, LLMs may generate explanations that appear coherent and convincing while being internally inconsistent or linguistically invalid. Such failures are especially problematic when they concern deeper linguistic dimensions that are not explicitly represented at the textual level.

One such dimension is phonetics. Despite good performance on simple phonological tasks such as rhyme detection or grapheme-to-phoneme conversion, LLMs exhibit significant weaknesses when explicit and structured phonetic reasoning is required. Tasks involving the manipulation, recomposition, and verification of sound segments remain challenging for current models.

Charades constitute a representative example of this type of reasoning. They rely on the phonetic decomposition of a target word into multiple segments, each indirectly evoked through lexical or semantic clues, before being recomposed to recover the final word. As such, charades can be viewed as *phonetic puzzles* whose resolution requires rigorous sound-level reasoning.

In practice, LLMs generally perform poorly when asked to generate or solve coherent phonetic charades. While they may sometimes detect that a solution is inconsistent, they are rarely able to precisely identify the phonetic source of the error or correct it. This observation motivates a systematic study of phonetic reasoning in LLMs and the development of targeted evaluation and improvement strategies.

1.2 Positioning with Respect to the State of the Art

Research on LLM reasoning capabilities has mainly focused on logical, arithmetic, and symbolic tasks, as well as abstract textual puzzles. In parallel, a limited number of benchmarks have evaluated the phonological abilities of LLMs, typically through atomic and weakly structured tasks such as syllable counting, rhyme recognition, or phonetic transcription.

To the best of our knowledge, no prior work has specifically investigated LLM performance on *structured phonetic puzzles* such as charades, nor proposed methods to improve phonetic reasoning in this setting. Moreover, no standardized dataset of phonetic charades currently exists for training or evaluating language models.

This lack of resources and analyses represents both a limitation of the current literature and an opportunity for research. Charades provide a particularly suitable experimental framework for studying phonetic reasoning, as they require explicit sound decomposition and allow fine-grained error analysis at each stage of the reasoning process.

1.3 Objectives and Contributions

The objective of this project, **P-CHARM** (*Phonetic Charade Reasoning Model for LLMs*), is to analyze and improve the phonetic reasoning capabilities of LLMs through the case study of charades. Specifically, we aim to:

- analyze the limitations of LLMs on phonetic puzzles;
- improve their performance using *few-shot learning* and *chain-of-thought prompting* to encourage explicit phonetic reasoning;
- introduce an original dataset of phonetic charades, addressing the absence of existing resources.

The dataset construction follows a semi-automatic combinatorial approach. Starting from a lexical dictionary, each target word is converted into its phonetic representation (IPA), segmented into phonetic units, and matched with existing words that correspond phonologically to each segment in order to form a coherent charade.

To objectively evaluate the quality of the generated charades, we propose an evaluation protocol based on phonetic similarity. The phonetic representations of the generated segments are concatenated and compared to the phonetic representation of the target word using phonological feature representations extracted with the PanPhon library, complemented by symbolic edit-distance measures. This yields a quantitative score reflecting global phonetic coherence.

The novelty of this work lies in the study of an underexplored form of reasoning in LLMs, the introduction of a new dataset of phonetic puzzles, and the evaluation of simple yet effective strategies to improve phonetic reasoning in language models.

2 Methodology

Our methodology is designed to analyze the phonetic reasoning limitations of LLMs and to evaluate simple strategies for improving their performance on phonetic puzzles. We focus on the task of phonetic charade generation, which requires explicit sound-level decomposition and recomposition.

2.1 Task Definition

The phonetic charade generation task can be formulated in two complementary settings. In the first setting, the model is provided with a target word and is asked to generate a corresponding phonetic charade by decomposing the word into sound-based segments. In the second setting, the model is allowed to freely select a target word and then construct a phonetic charade for the chosen word.

In both cases, the generated charade must satisfy two constraints: (i) the phonetic concatenation of the proposed segment words must match the phonetic form of the target word, and (ii) each segment must be associated with a semantically coherent clue. These formulations allow us to evaluate both controlled phonetic reasoning and the model's ability to autonomously select suitable targets.

This task explicitly requires reasoning over phonetic representations rather than relying solely on surface-level textual similarity.

2.2 Prompting Strategies

We compare two prompting configurations to analyze their impact on phonetic reasoning:

- **Base prompting**, where the LLM is directly asked to generate a phonetic charade without additional guidance.
- **Guided prompting**, which combines few-shot learning and chain-of-thought prompting. In this configuration, the model is provided with a small number of correctly solved phonetic charades as examples and is explicitly encouraged to reason step by step, including phonetic transcription, segmentation, and recomposition.

The goal of guided prompting is to force the model to externalize its phonetic reasoning process and to reduce silent phonetic inconsistencies that commonly arise in direct generation.

3 Phonetic Charade Dataset

3.1 Dataset Motivation

As no standardized dataset of phonetic charades currently exists, we propose the creation of a new dataset specifically designed to support both evaluation and potential training of LLMs on phonetic reasoning tasks. The dataset aims to provide explicit phonetic supervision and controlled phonological structure.

3.2 Dataset Construction

The dataset is constructed using a semi-automatic combinatorial pipeline. First, a target word is selected from a lexical dictionary. The word is then converted into its phonetic representation using the International Phonetic Alphabet (IPA). This phonetic form is segmented into valid phonetic sub-units.

For each phonetic segment, existing words whose phonetic realization matches the segment are searched within the dictionary. These words constitute the phonetic components of the charade. This procedure guarantees that the resulting decompositions are phonologically valid by construction, while allowing the generation of a large number of diverse examples.

The dataset deliberately focuses on the phonetic structure of charades rather than on fully specified natural language clues, which can be generated or inferred by a language model during prompting.

3.3 Dataset Structure

Each dataset entry contains:

- a target word and its phonetic transcription;
- one or several valid phonetic segmentations;
- the corresponding segment words for each segmentation;
- the reconstructed phonetic form obtained by concatenation.

The dataset is designed to be easily extensible and applicable to other languages with available phonetic resources.

4 Evaluation Metrics and Protocol

4.1 Phonetic Similarity Metric

To objectively evaluate the quality of generated charades, we propose a phonetic-based evaluation metric. The phonetic transcriptions of the segment words generated by the model are concatenated and compared to the phonetic transcription of the target word.

Similarity is computed using vectorial representations of phonological features derived from the PanPhon library. The distance between the generated phonetic feature sequence and the target feature sequence provides a quantitative score reflecting global phonetic coherence. In addition, the Levenshtein distance between phonetic strings is computed as a complementary symbolic measure of phonetic similarity.

4.2 Evaluation Protocol

For each target word, charades generated under different prompting strategies are evaluated using the proposed phonetic similarity metrics. This enables a direct comparison between base prompting and guided prompting configurations.

In addition to quantitative phonetic scores, qualitative analyses are conducted to identify recurrent error patterns, such as incorrect phonetic segmentation, approximate phonetic matches, or valid phonetic constructions paired with semantically implausible segment choices.

This evaluation protocol is designed to assess phonetic correctness and reasoning transparency, without relying on surface-level textual overlap.

5 Results

We evaluated the phonetic charade generation task using several large language models under different prompting configurations. Experiments were primarily conducted using Google Gemini models (Gemini 2.5 Flash Lite and Gemini 2.5 Flash) through an application interface requiring structured JSON outputs.

5.1 *Impact of Output Formatting Constraints*

An important experimental observation emerged regarding output formatting. When models were directly instructed to produce structured JSON outputs, the inclusion of an example JSON format acted as an implicit form of guidance. This constraint alone led to noticeable improvements in phonetic coherence, even under base prompting conditions.

As a result, the performance gap between base prompting and guided prompting was reduced when JSON formatting was imposed from the start. To mitigate this unintended bias, we adopted a two-step prompting strategy: the model was first asked to generate a charade in free text form, and a second prompt was then used to convert the output into the required JSON format. This approach allowed us to better isolate the effect of phonetic reasoning guidance from formatting-induced improvements.

5.2 *Quantitative Evaluation and Qualitative Analysis*

Quantitative evaluation results indicate that guided prompting consistently improves phonetic coherence compared to base prompting. Lower values for both score and distance indicate superior performance, with 0 being the optimal result. On average, guided prompting achieved a phonetic similarity score of 0.75, compared to 3.46 for base prompting, as illustrated in Figure 1. Similarly, the average Levenshtein distance decreased from 7.83 to 2.67 under guided prompting.

These improvements were observed across both Gemini 2.5 Flash Lite and Gemini 2.5 Flash models, with Gemini 2.5 Flash achieving the highest overall performance. However, the magnitude of improvement varied depending on the target word and its phonetic decomposition.

Beyond quantitative scores, an important qualitative difference emerged between prompting configurations. Under base prompting, models frequently generated non-existent or morphologically invalid words. While such invented forms could sometimes yield phonetic sequences relatively close to the target, they remain linguistically invalid and undermine the coherence of the generated charade. In contrast, guided prompting strongly constrained the model toward producing existing lexical items. Even when phonetic reconstruction was imperfect, the resulting charades remained composed of valid words, leading to globally more coherent and interpretable outputs.

Qualitative analysis revealed that base prompting frequently led to incorrect phonetic segmentations, approximate phonetic matches, or segment choices that failed to reconstruct the target phonetic form. In contrast, guided prompting encouraged more explicit reasoning steps, resulting in segmentations that more closely aligned with the target phonetic structure.

Nevertheless, performance remained sensitive to the phonetic segmentation of the target word. Words with clear and well-aligned phonetic sub-units were more reliably handled by the models, whereas ambiguous or less frequent phonetic decompositions continued to pose challenges, even under guided prompting. For instance, long and morphologically complex words such as *hippopotame* proved particularly challenging. These words admit multiple plausible phonetic segmentations, many of which do not correspond to frequent or semantically salient lexical items. In such cases, models often produced segmentations that were locally plausible but globally inconsistent when recomposed. Conversely, words with clearer phonetic boundaries, such as *enfant*, were handled more reliably, especially under guided prompting.

5.3 *Cross-Model Observations*

Additional manual experiments were conducted with other LLMs, including ChatGPT4 and Claude AI, using interactive interfaces without API access. While these models exhibited similar qualitative trends, namely poor baseline performance and improvements under guided prompting, these results are reported qualitatively due to the lack of controlled, automated evaluation.

6 Discussion and Limitations

The experiments conducted in this project highlight several important observations regarding phonetic reasoning in Large Language Models. While LLMs are capable of producing fluent and coherent textual explanations, they exhibit clear weaknesses when explicit manipulation of phonetic representations is required. This is particularly evident in the task of charade generation, where phonetic segmentation and recomposition must be jointly satisfied.

Our analysis suggests that prompting strategies play a crucial role in mitigating these limitations. Few-shot learning and chain-of-thought prompting consistently lead to more phonologically coherent charades than direct prompting. By encouraging the model to explicitly reason about phonetic transcriptions and intermediate segments, these methods reduce silent phonetic inconsistencies and improve reasoning transparency. This supports the hypothesis that LLMs benefit from structured guidance when operating on non-textual linguistic dimensions.

It is important to emphasize that guided prompting does not fully solve the phonetic reasoning problem. Even with explicit reasoning steps and phonetic supervision, models continue to struggle with complex or ambiguous target words, particularly those involving long phonetic sequences, rare segmentations, or morphologically dense constructions. In such cases, guided prompting often fails to produce a perfectly valid decomposition.

Nevertheless, guided prompting consistently yields outputs that are more coherent than those produced under base prompting. While errors persist, they tend to involve incomplete or suboptimal segmentations rather than uncontrolled phonetic drift or invented lexical items. This indicates that guided prompting improves the stability and reliability of phonetic reasoning, even when it does not fully succeed.

A natural question raised during this project concerns the necessity of using LLMs at all, given that a combinatorial, rule-based program can generate phonologically valid charades. While such a program is effective for dataset construction and guarantees phonetic correctness, it lacks several essential properties. In particular, it does not account for semantic relevance, usage frequency, or contextual appropriateness of the generated words. As a result, it may produce charades involving rare, unnatural, or pragmatically irrelevant words.

In contrast, LLMs are able to adapt charade generation to semantic context, thematic constraints, or target audiences (e.g., educational level or domain). More importantly, LLMs provide explicit reasoning steps and explanations, which are central to the study of reasoning abilities. In this sense, the combinatorial program and the LLM serve complementary roles: the former enables controlled data generation, while the latter enables reasoning, generalization, and interaction.

Despite these promising observations, our work presents several limitations. First, no model was explicitly trained or fine-tuned on the proposed dataset due to time constraints; our analysis is therefore restricted to prompting-based improvements. Second, the evaluation relies on phonetic embeddings, which, while informative, may not capture all aspects of human phonological perception. Third, the study focuses on a single type of phonetic puzzle and a single language, limiting the scope of generalization. Moreover, only a subset of models was evaluated through the application interface, due to API accessibility constraints. Other models were assessed qualitatively through manual interaction, which limits direct quantitative comparison.

These limitations nevertheless define clear directions for future research.

7 Conclusion and Perspectives

In this project, we investigated the phonetic reasoning capabilities of Large Language Models through the case study of charades, a class of structured phonetic puzzles that require explicit sound-level manipulation. We showed that, despite their strong performance on textual reasoning tasks, LLMs struggle with phonetic coherence when reasoning is not explicitly guided.

We introduced a novel phonetic charade dataset constructed through a semi-automatic combinatorial pipeline, addressing the absence of existing resources for this task. We further proposed an evaluation protocol based on phonetic similarity using phonological feature representations derived from PanPhon, complemented by symbolic edit-distance measures. This enables a quantitative assessment of phonetic coherence. Our analysis indicates that simple prompting strategies, such as few-shot learning combined with chain-of-thought prompting, can significantly improve phonetic reasoning performance without modifying model parameters.

This work opens several promising perspectives. A natural next step would be to train or fine-tune LLMs on the proposed dataset to assess whether phonetic reasoning can be learned more robustly and generalized beyond prompting. The dataset could also be extended to other languages and phonological systems, enabling cross-linguistic studies of phonetic reasoning. Another important direction is to evaluate whether improvements gained on charades transfer to other phonetic puzzles, such as riddles, puns, or rhyme-based tasks, thereby testing the generalization of phonetic reasoning skills. From an application perspective, the evaluation framework could be extended to support multiple LLM backends in a unified and automated manner, enabling large-scale and reproducible comparisons across architectures.

More broadly, this project highlights the importance of evaluating and developing LLMs along deeper linguistic dimensions beyond text-level reasoning. Phonetic puzzles provide a controlled and interpretable framework for studying sound-based reasoning, and we hope this work will encourage further research in this direction.

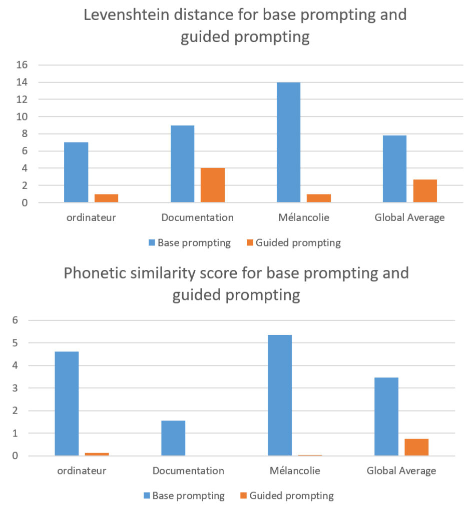

Figure 1. Examples and Average phonetic similarity score and Levenshtein distance for base prompting and guided prompting across Gemini models.

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